Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 2 Quiz 1 - Kinematics in Two Dimensions

Mock Exam B · Form 2: Single-step application

Multiple Choice Questions: Select the best answer. (1 point each)

1. A ball is launched at 15 m/s, 40° above horizontal, and lands at its launch height. Ignore drag. Find total speed at the highest point. Use g = 9.8 m/s².

(A) 11.491 m/s

(B) -11.491 m/s

(C) 22.981 m/s

(D) 5.745 m/s

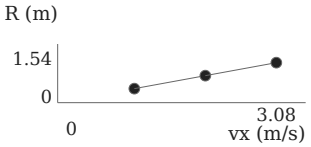
2. A ball launches horizontally from a 1.25 m table. Use g=10 m/s² and neglect drag. vx (m/s) -> R (m): (1.08, 0.54); (2.08, 1.04); (3.08, 1.54). What is the slope between the endpoint measurements?

(A) 0.3

(B) 0.5

(C) 0.9

(D) 0.7



3. In a horizontal-launch lab, predicted range is 1.2 m and mean measured range is 0.984 m. What is the percent error relative to the prediction?

(A) 36 %

(B) 9 %

(C) 18 %

(D) -18 %

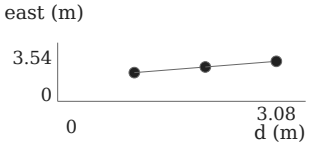
4. A map route starts at east coordinate 2 m and runs at 60° north of east. d (m) -> east (m): (1.08, 2.54); (2.08, 3.04); (3.08, 3.54). Which relation models this apparatus?

(A) Y=-0.5X+(2)

(B) Y=1X+(2)

(C) east=e0+d cosθ

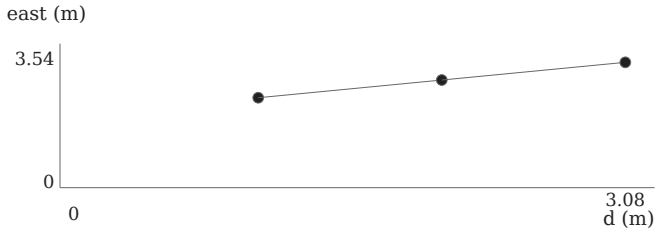
(D) Y=0.5X²+(2)



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A map route starts at east coordinate 2 m and runs at 60° north of east. d (m) -> east (m): (1.08, 2.54); (2.08, 3.04); (3.08, 3.54).

a. For the eastward projection setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Label the axes and units and demonstrate interpolation between two readings. **(3 pt)**

6. Launch at 30° and land at launch height, without drag. Use g=10 m/s². u² (m²/s²) -> R (m): (1.08, 0.09353); (2.08, 0.1801); (3.08, 0.2667).

a. Build the model for level-ground range from physics and predict both slope and intercept. **(2 pt)**

b. Label the axes and units and demonstrate interpolation between two readings. **(2 pt)**

7. A ball launches horizontally from a 1.25 m table. Use g=10 m/s² and neglect drag. vx (m/s) -> R (m): (1.08, 0.54); (2.08, 1.04); (3.08, 1.54). Build the model for horizontal-launch target from physics and predict both slope and intercept. **(3 pt)**